

Topic: Michelson interferometer, Diffraction of light, Polarization of light

Michelson interferometer:

An interferometer is a device that can be used to measure lengths or changes in length with great accuracy by means of interference fringes. We describe the form originally devised and built by A. A. Michelson in 1881.

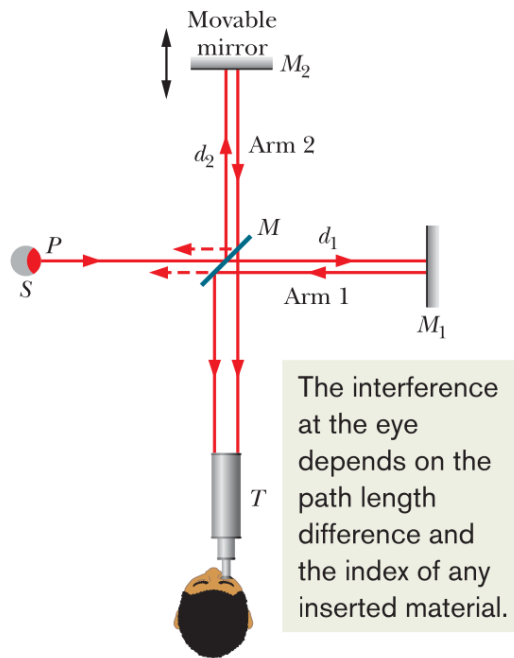


Figure 1: Michelson's interferometer, showing the path of light originating at point P of an extended source S . Mirror M splits the light into two beams, which reflect from mirrors M_1 and M_2 back to M and then to telescope T . In the telescope an observer sees a pattern of interference fringes.

Working Principle

YOUTUBE LINK : <https://youtu.be/ffBcKYnjclc?si=o6uafZv8d-0PjmrP> Con-

sider light that leaves point P on an extended source S and encounters beam splitter M . A beam splitter is a mirror that transmits half the incident light and reflects the other half. For convenience, we assume that this mirror possesses negligible thickness. At M , the light thus divides into two waves. One proceeds by transmission toward mirror M_1 at the end of one arm of the instrument; the other proceeds by reflection toward mirror M_2 at the end of the other arm. The waves are entirely reflected at these mirrors and are sent back along their directions of incidence, each wave eventually entering telescope T . The observer sees a pattern of curved or approximately straight interference fringes; in the latter case, the fringes resemble the stripes on a zebra.

Mirror Shift

The path length difference for the two waves when they recombine at the telescope is $2d_2 - 2d_1$, and anything that changes this path length difference will cause a change in the phase difference between these two waves.

Insertion of a Material

A shift in the fringe pattern can also be caused by the insertion of a thin transparent material into the optical path of one of the mirrors.

Diffraction of Light

Diffraction is the bending of light waves around obstacles and the spreading of light waves when they pass through small apertures. It occurs when *the size of the obstacle or aperture is comparable to the wavelength of the light*.

Types of Diffraction

Diffraction of light can be classified into two types:

1. Fresnel Diffraction

Fresnel diffraction occurs when the source of light or the screen (or both) are at a **finite distance** from the diffracting aperture or obstacle. The wavefronts are spherical or cylindrical in this case.

Characteristics of Fresnel Diffraction:

- The source and screen are at a finite distance.
- The wavefronts are either spherical or cylindrical.
- It is observed in cases such as the diffraction at a circular aperture or an opaque disc.

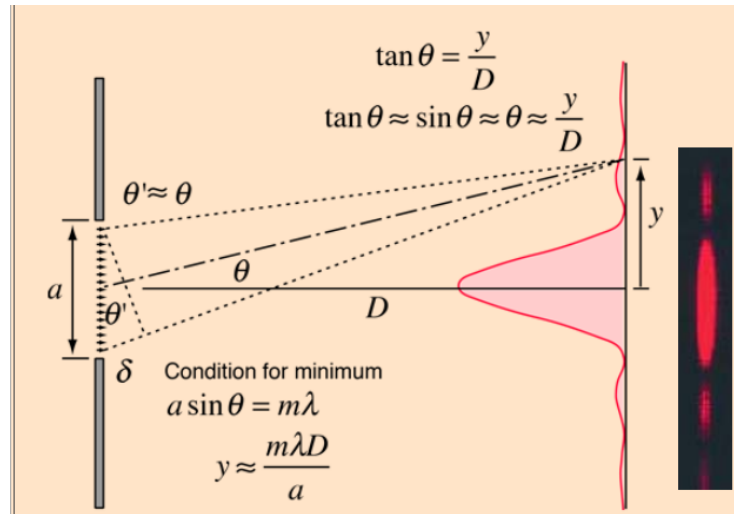
2. Fraunhofer Diffraction

Fraunhofer diffraction occurs when the source of light and the screen are at **infinite distances** from the diffracting aperture or obstacle, making the incident wavefronts parallel.

Characteristics of Fraunhofer Diffraction:

- The source and screen are at infinite distances.
- The wavefronts are plane.
- The diffraction pattern is sharper and well-defined.
- Examples include diffraction due to a single slit, double slit, or a diffraction grating.

Single-Slit Diffraction



When a monochromatic light wave passes through a single narrow slit of width a , it produces a central bright fringe with alternating dark and bright fringes on either side. The condition for minima (dark fringes) is given by:

$$a \sin \theta = m \lambda, \quad m = \pm 1, \pm 2, \pm 3, \dots$$

where λ is the wavelength of light and m is the order of diffraction.

The central maximum is the brightest and widest, while the intensity of secondary maxima decreases as we move away from the center.

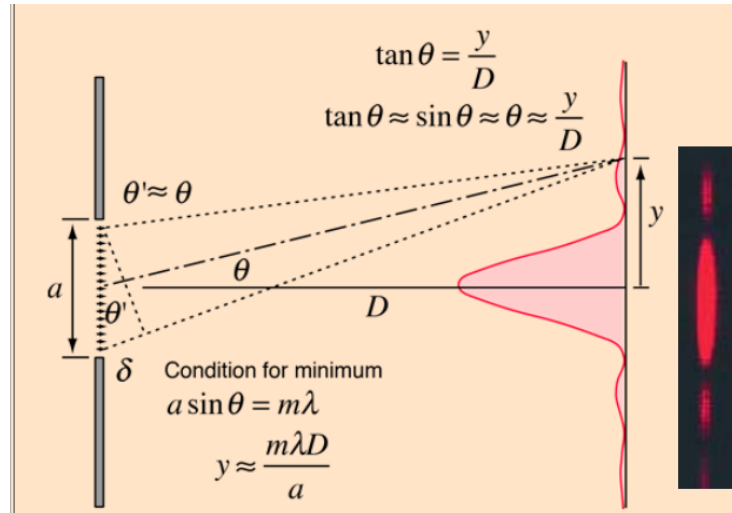
PROBLEMS ON DIFFRACTION

Problem: 1

The distance between the first and fifth minima of a single-slit diffraction pattern is 0.35 mm with the screen 40 cm away from the slit, when light of wavelength 550 nm is used.

1. Find the slit width (a).
2. Calculate the angle θ_1 of the first diffraction minimum.

Solution:



(a) Finding the Slit Width:

The distance between the first ($m = 1$) and fifth ($m = 5$) minima on the screen is related to the slit width by the formula:

$$y_5 - y_1 = \frac{5\lambda L}{a} - \frac{\lambda L}{a} = \frac{(5 - 1)\lambda L}{a} = \frac{4\lambda L}{a}.$$

Rearranging for a , we have:

$$a = \frac{4\lambda L}{y_5 - y_1}.$$

Substitute the given values:

$$\begin{aligned}\lambda &= 550 = 550 \times 10^{-9} \text{ m}, \\ L &= 40 = 0.40 \text{ m}, \\ y_5 - y_1 &= 0.35 = 0.35 \times 10^{-3} \text{ m}.\end{aligned}$$

Now calculate:

$$a = \frac{4 \times (550 \times 10^{-9}) \times 0.40}{0.35 \times 10^{-3}} = \frac{8.8 \times 10^{-7}}{0.35 \times 10^{-3}} = \frac{8.8}{0.35} \times 10^{-4} = 25.14 \times 10^{-4} \text{ m} = 2.51.$$

Thus, the slit width is:

$$\boxed{a = 2.51}.$$

(b) Calculating the Angle θ_1 :

The angle for the first diffraction minimum is given by:

$$a \sin \theta_1 = m\lambda.$$

For the first minimum ($m = 1$):

$$\sin \theta_1 = \frac{\lambda}{a}.$$

Substitute the values:

$$\sin \theta_1 = \frac{550 \times 10^{-9}}{2.51 \times 10^{-3}} = 2.19 \times 10^{-4}.$$

Now calculate θ_1 :

$$\theta_1 \approx 0.0125^\circ.$$

Thus, the angle for the first minimum is:

$$\boxed{\theta_1 \approx 0.0125^\circ}.$$

Problem: 2

What must be the ratio of the slit width (a) to the wavelength (λ) for a single slit to have the first diffraction minimum at $\theta = 45^\circ$?

Solution:

The condition for the m -th diffraction minimum in a single-slit diffraction pattern is given by:

$$a \sin \theta = m\lambda.$$

For the first diffraction minimum ($m = 1$), this becomes:

$$a \sin \theta = \lambda.$$

Rearranging for the ratio a/λ :

$$\frac{a}{\lambda} = \frac{1}{\sin \theta}.$$

Substitute $\theta = 45^\circ$:

$$\sin 45^\circ = \frac{\sqrt{2}}{2}.$$

Thus:

$$\frac{a}{\lambda} = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}.$$

Numerically:

$$\sqrt{2} \approx 1.414.$$

Therefore, the required ratio is:

$$\boxed{\frac{a}{\lambda} \approx 1.414}.$$

Problem: 3

A plane wave of wavelength 590 nm is incident on a slit with a width of $a = 0.40 \text{ mm}$. A thin converging lens of focal length 70 cm is placed between the slit and a viewing screen and focuses the light on the screen.

- (a) How far is the screen from the lens?
- (b) What is the distance on the screen from the center of the diffraction pattern to the first minimum?

Solution:

(a) Distance from the Lens to the Screen:

The distance from the lens to the screen is simply the focal length of the lens since it focuses parallel rays (plane wave) onto the screen. Thus:

$$\text{Distance from lens to screen} = f = 70\text{cm}.$$

Distance = 70cm

(b) Distance from the Center of the Diffraction Pattern to the First Minimum:

The position of the first minimum in the single-slit diffraction pattern is given by the formula:

$$y_1 = \frac{\lambda f}{a},$$

where:

$$\lambda = 590 = 590 \times 10^{-9} \text{ m},$$

$$f = 70 = 0.70 \text{ m},$$

$$a = 0.40 = 0.40 \times 10^{-3} \text{ m}.$$

Substitute these values into the formula:

$$y_1 = \frac{(590 \times 10^{-9}) \times 0.70}{0.40 \times 10^{-3}}.$$

Simplify:

$$y_1 = \frac{4.13 \times 10^{-7}}{0.40 \times 10^{-3}} = 1.0325 \times 10^{-3} \text{ m} = 1.03.$$

Thus, the distance from the center to the first minimum is:

$y_1 = 1.03$.

Problem : 4

Monochromatic light of wavelength $\lambda = 441 \text{ nm}$ is incident on a narrow slit. On a screen 2.00 m away, the distance between the second diffraction minimum and the central maximum is 1.50 cm .

(a) Calculate the angle of diffraction θ of the second minimum.

(b) Find the width of the slit.

Solution:

For single-slit diffraction, the condition for the minima is given by:

$$a \sin \theta = m\lambda$$

where: - a is the width of the slit, - θ is the diffraction angle, - m is the order of the minimum (for the second minimum, $m = 2$), - λ is the wavelength of the light.

Given: $\lambda = 441 \text{ nm} = 441 \times 10^{-9} \text{ m}$, The screen distance $L = 2.00 \text{ m}$, The distance from the central maximum to the second diffraction minimum $y_2 = 1.50 \text{ cm} = 0.015 \text{ m}$.

We can approximate the angle θ using the small angle approximation, where $\sin \theta \approx \tan \theta \approx \frac{y_2}{L}$. Thus:

$$\tan \theta = \frac{y_2}{L} = \frac{0.015 \text{ m}}{2.00 \text{ m}} = 7.5 \times 10^{-3}$$

Since θ is small, we have:

$$\theta \approx 0.430, \text{ deg}$$

Now, **for part (b)**, we use the diffraction condition for the second minimum ($m = 2$):

$$a \sin \theta = 2\lambda$$

Substituting $\sin \theta \approx \theta$ for small angles:

$$a \cdot 7.5 \times 10^{-3} = 2 \times 441 \times 10^{-9}$$

Solving for a :

$$a = \frac{2 \times 441 \times 10^{-9}}{7.5 \times 10^{-3}} = 1.18 \times 10^{-4} \text{ m} = 0.118 \text{ mm}$$

Thus, the width of the slit is $a = 0.118 \text{ mm}$.

Problem : 5

For what value of the slit width a will the first minimum for red light of wavelength $\lambda = 650 \text{ nm}$ appear at an angle $\theta = 15^\circ$?

Solution

In single-slit diffraction, the condition for the first minimum is given by:

$$a \sin \theta = m\lambda,$$

where:

- a is the slit width,
- θ is the angle of the first minimum,
- λ is the wavelength of the light,
- m is the order of the minimum (for the first minimum, $m = 1$).

Substituting the given values:

$$a \sin 15^\circ = 1 \cdot 650.$$

Solve for a :

$$a = \frac{\lambda}{\sin \theta}.$$

Convert λ to meters:

$$\lambda = 650 \times 10^{-9} \text{ m}.$$

Substitute the values:

$$a = \frac{650 \times 10^{-9}}{\sin 15^\circ}.$$

Using $\sin 15^\circ \approx 0.2588$:

$$a = \frac{650 \times 10^{-9}}{0.2588}.$$

Calculate a :

$$a \approx 2.51 \times 10^{-6} \text{ m}$$

Difference Between Interference and Diffraction

Both interference and diffraction are wave phenomena related to the superposition of waves, but they have distinct differences. The key differences are outlined below:

- **Interference** occurs due to the superposition of two or more coherent waves. It requires at least two wave sources with a constant phase relationship. The result is an alternating pattern of bright and dark fringes with **uniform spacing**, such as in Young's double-slit experiment.
- **Diffraction** occurs when a wave encounters an obstacle or passes through a narrow aperture, causing it to bend and spread. Unlike interference, diffraction can occur with a single wave source. The resulting pattern typically has a central bright fringe that is wider, with **subsequent fringes of decreasing intensity**, as seen in single-slit diffraction or diffraction gratings.

Polarization

What Is Polarization?

Light is the interaction of electric and magnetic fields travelling through space. The electric and magnetic vibrations of a light wave occur perpendicularly to each other. The electric field moves in one direction and the magnetic field in another 'perpendicular to each other. So, we have one plane occupied by an electric field, another plane of the magnetic field perpendicular to it, and the direction of travel is perpendicular to both. These electric and magnetic vibrations can occur in numerous planes.

A light wave that is vibrating in more than one plane is known as unpolarized light. The light emitted by the sun, by a lamp or a tube light are all unpolarized light sources. As you can see in the image below, the direction of propagation is constant, but the planes on which the amplitude occurs are changing.

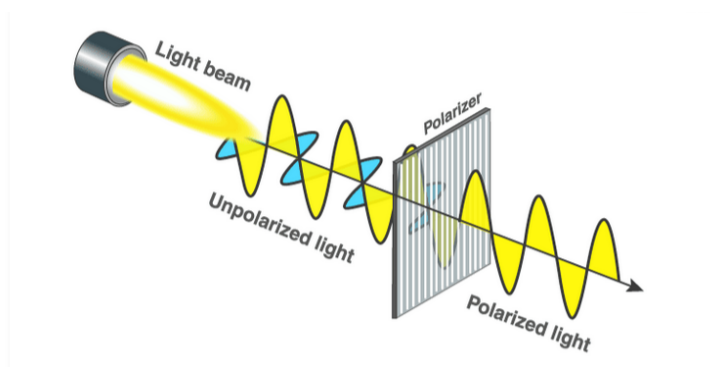


Figure 2: Polarization

The other kind of wave is a polarized wave. Polarized waves are light waves in which the vibrations occur in a single plane. Plane polarized light consists of waves in which the direction of vibration is the same for all waves.

The process of transforming unpolarized light into polarized light is known as polarization

Types of Polarization

Following are the three types of polarization depending on how the electric field is oriented:

- **Linear polarization** : In linear polarization, the electric field of light is limited to a single plane along the direction of propagation.
- **Circular polarization** : The propagation of the occurring electric field will be in a circular motion. There are two linear components in the electric field of light that are perpendicular to each other such that their amplitudes are equal, but the phase difference is $\pi/2$.
- **Elliptical polarization** : The electric field of light follows an elliptical propagation.

Polarization Applications

- Polarization is used in sunglasses
- Polarization is used for differentiating between transverse and longitudinal waves.
- Three-dimensional movies